MIL-T-55107(SigC)
8 July 1960

MILITARY
SPECIFICATION

TRAINER, RADAR AN/ULT-T4( ).

1. SCOPE

1.1 This specification covers one type of L-Band radar signal interference trainer, designated as Trainer, Radar AN/ULT-T4( ), capable of simulating and transmitting various types of interference signals for use in the training of radar operators.

2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on the date of the invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATION

Military

MIL-C-3885	Cable assemblies and cord assemblies, elec.(for use in elec. comm. and assoc. equip.)
MIL-P-11268	Parts, materials and processes.
MIL-I-11748	Interference reduction for elect. equip.
MIL-M-13231	Marking of electronic items.
MIL-F-14072	Finishes for ground signal equipment.

STANDARDS

Military

MIL-STD-105	Sampling for expensive testing by attributes.
MIL-STD-169	Extreme temperature cycle.
MIL-STD-170	Moisture resistance test cycle for ground signal equipment.
MIL-STD-252	Wired equipment, classification of visual and mechanical defects.

DRAWINGS

Signal Corps

SC- A - 362100	Packaging and packing design illustrations and bills of material for preproduction equipment, preparation of.

FSC 6940

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title or number and symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Trainer, Radar AN/ULT-T4( ) shall be capable of transmitting low power, unmodulated continuous wave (CW), pulsed, sine wave, square wave, and noise modulated CW over a frequency range of at least 1220 to 1350 megacycles per second. The equipment shall operate as a swept frequency or as a spot jammer.

3.2. Parts, materials and processes; general.- In addition to the requirements of this specification the requirements of Specification MIL-P-11268, including the selection requirements therein, shall apply.

3.3 Procurement model.- A procurement model of the equipment will be available for inspection by prospective bidders and will be lent to the contractor.

3.3.1 Construction.- Unless otherwise specified herein, or in the invitation for bids, physical construction of the equipment shall conform to the model. In case of conflict between the specified performance characteristics for the equipment and the performance of the model the specified performance characteristics shall govern.

3.3.1.1 Size and weight.- The size of Trainer, Radar AN/ULT-T4( ) shall not exceed that of the model and the weight shall not exceed 100 lbs.

3.3.1.2 Material cases.- The cases shall be constructed of aluminum or a three or more ply material, phenolic-resin-bonded, having the outer plies of aluminum and the inner ply or plies of a suitable material.

3.3.1.3 Antenna.- The antenna shall be a modified Yagi Array with a gain of 11 ± 2 db.

3.3.1.4 Transit case.- The transit case shall provide storage space for all accessories and running spare parts, and shall protect the equipment against damage during transit and storage.

3.4. Preproduction samples.- The contractor shall furnish preproduction samples for approval, if required by the invitation for bids and the contract. (See 4.3 and 6.2(c)).

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3.5 Finish, protective.- The equipment shall be given a protective finish in accordance with Specification MIL-F-14072. This includes the finish of hardware, such as handles, hinges, screws, etc., and the necessary touch up after mounting. The final paint film on type I surfaces shall be green color (Olive Drab), semi-gloss enamel, matching a color chip provided by the procuring agency(See 6.3 and 4.4.)

3.6. Marking.- Marking shall conform to Specification MIL-M-13231 (See 4.4). Front panel marking shall be group I as described in that specification.

3.6.1 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.6.2 Serial numbers.- Each Trainer, Radar AN/ULT-T4 shall be serial numbered.

3.7 Interchangeability.- Like Units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment.(See 4.7). Individual items shall not be hand-picked for fit or performance, however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.8 Service conditions.- The equipment shall meet the following service conditions:

3.8.1 Operation.- Continuous use for a period of 400 hours under the worldwide environmental conditions specified in 3.8.2 through 3.8.10, with a duty cycle of eight(8) hours on and eight(8) hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample (See 3.4) shall be considered as compliance with this requirement.

3.8.2 Temperature (See 4.8.1).-

(a) Operating.- Ambient temperature in the range of +150° F to -40° F, (The 150° F temperature includes the effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours at any one time.

(b) Nonoperating.- Exposure in the range of +160°F to -80°F, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time.

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3.8.3 Relative humidity(See 4.8.2).- Up to 97 percent relative humidity for 20 hours and exposure at 100 percent relative humidity, with condensation, for one(1) hour.

3.8.4 Elevation(See 4.8.3).-

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.8.5 Immersion (See 4.8.4).- Three feet of water for two(2) hours. (In transit case).

3.8.6 Orientation:

(a) Operating: Any orientation up to 20 degrees from normal operating position(that is: forward,backward, left, or right).

(b) Nonoperating: Storage in any position for a period of two years. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample(See 3.4) shall be considered as compliance with this requirement.

3.8.7 Rain (See 4.8.5).- Two inches of rainfall per hour, with wind at 20 miles per hour.

3.8.8 Sand and Dust(See 4.8.6).- As encountered during heavy traffic over dry clay roads or in desert areas.

3.8.9 Vibration (See 4.8.7).- With the equipment mounted in its normal operating position the amplitude of vibration of any part subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second.

3.8.10 Bounce and shock.- The equipment shall be capable of meeting the bounce, bench and drop tests of 4.8.8, 4.8.9 and 4.8.10. There shall be no physical damage other than minor surface abrasions and the equipment shall meet the operation requirements of 3.11.

3.9. Suppression of radio interference.- The equipment shall meet the emanation and susceptibility limits of Specification MIL-I-11748 for Class Ib. (See 4.4). Application of suppression components and techniques shall conform tothat specification. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance with the specified emanation and susceptibility limits he shall obtain approval from the contracting officer, before proceeding with the modification.

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3.10 Preconditioning.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6 (Also see 4.5)

3.11 Operation requirements.-

3.11.1 Input power.- The unit shall be capable of meeting all the requirements of this specification, without readjustment, when operating from a power source of 105 tol20 volts; 48 to 62 cycles and 390 to 440 cycles, single phase.

3.11.1.1 Power switch.- The power switch shall be a three-position, rotary switch with position from left to right as follows: OFF- STANDBY- TRANSMIT. In the STANDBY position voltage shall be applied only to the electron tube filaments. In the TRANSMIT position the equipment shall be capable of providing all required functions.

3.11.2 Power supply.- The power supply shall supply the following voltages when operated from a 105 to 120 volts AC, 48 to 62 cycle per second, or 390 to 450 cycle per second, single phase source;

+ 300 ± 5 volts dc, regulated, at 160 ma.

+ 28 ± 3 volts dc, unregulated @ 100 ma.

6.3 ± 0.3 volts ac, unregulated @ 9.00 amps.

3.11.2.1 Fusing.- Both sides of the input power leads shall be fused between the input power plug and the power switch(See 4.9.3.1).

3.11.3 R-F stage.- The R-F stage shall be tunable over a frequency range of 1220 to 1350 mc and shall provide an average power output of at least 2 milliwatts, without attennation, into a 50 ohm matched load. The power output shall not vary more than ±2 db over the frequency range.

3.11.3.1 Sweep frequency.- Automatic R-F frequency sweeping shall be accomplished in the R-F stage with the sweep sectors being 50 mc and 130 mc wide. The rate of sweeping shall be adjustable from 0 to 6 sweeps per minute in discrete unit steps. The sweep sector widths and sweep rates shall be calibrated within ± 20%. The center frequency of the 50 mc sector shall be continuously variable from 1220 to 1350 mc and shall be manually selected.

3.11.3.2 R-F output attennation.- The R-F power output shall be attennated in steps of 0,1, 2, 3, 5, 10, 20 and 41 db.

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3.11.4 Modulation.- The unit shall supply any of the following R-F outputs, as selected by the operator:

(a) Unmodulated.

(b) Sine wave, variable from 15 cps to 15 kcs, and fixed at 50 kcs and 100 kcs; measured frequencies to be within ± 20% of dial calibration. The amplitude of modulation shall not vary more than ± 10%.

(c) Square wave, at same conditions as the sine wave modulation.

(d) Pulse modulation at rep rates of 15 cps to 15 kcs, 50 kcs and 100 kcs. The pulse width shall be continuously variable from one(1) microsecond to 5 microseconds. The dial calibration shall be accurate (PRF and pulse width) to ± 20% and the amplitude shall be stable within ± 10%.

(e) Noise modulated in amplitude with at least a 2 mc bandwidth, variable in amplitude from 0 to 100 %.

(f) Shall be capable of being modulated from an external source.

3.11.5 Antenna gain.- The antenna shall have a gain of 11 ± 2 db over the frequency range of 1220 to 1350 mc.

3.11.6 Audio power output.- The audio amplifier shall provide an audio level of at least 300 milliwatts maximum across 600 ohms and shall be variable to provide a comfortable listening level in the headset.

3.11.7 Attennator pad,28 db.- The 28 db attennator pad shall be Douglas Microwave Co., Model 126A, or equal.

3.11.8 Cable assemblies.- Power and audio frequency cable and cord assemblies used for voltages up to and including 600 volts ac or corresponding dc shall conform to all the requirements of Specification MIL-C-3885. Cable assemblies used for over 600 volts ac or equivalent dc shall be subject to approval by the contracting offices.

3.12 Louvers.- The louvers in the equipment case shall be covered internally with an approved metal screening electrically shorted to the case to keep spurious emanations within the limits prescribed by Specification MIL-I-11748.

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3.13 Technical literature, tools and running spare parts.- Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order(See 6.2(f) and(g)).

3.14 Workmanship.-The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

(a) In specification MIL-P-11268:

General requirements for plastic material and plastic parts.	Process for soldering cleaning.
Wiring and cabling, including:	Riveting
Slack	General requirements for securing of parts.
Protection	Cleaning of welding
Clearance	Process for welding
Splicing	Mechanical operation of controls.
Connections, general	Gears
Grounding, general	Self tapping screws.
Shielding on wire and cable	

Flux and cleaning agents for soldering.

(b) In this specification:

3.2 Parts, materials and processes
3.3.1 Construction
3. 3. 1.4 Transit case (Storage)
3.5 Finish, protective
3.6 Marking
3.12 Louvers
3.13 Technical literature, tools and running spare parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility.- The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2. Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Acceptance inspection.
(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5)

(2) Acceptance inspection of preparation for delivery (See 4.11)

4.3. Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering the items listed in 4.4 and inspection specified for group A, group B and group C (See tables II, III, and IV, respectively); The preproduction inspection will normally be performed in this order: (1) Vibration, (2) bounce, (3) shock, bench handling, (4) shock, drop, and (5) immersion; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I. - Preproduction inspection

INSPECTION (For additional preproduction inspection see 4.3)	Rqt. Par.	Insp Par.
Service conditions.	3.9	4.8

4.4. Acceptance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Finish.	3.5
Marking.	3.6
Suppression, R-F.	3.9
Cable assemblies.	3.11.8

4.5 Acceptance inspection of equipment before preparation for delivery.- The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4. and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government - - - at its discretion - - - may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements.(See 6.7). Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to Group A, Group B, or Group C inspection shall be preconditioned after final assembly.(See 3.10)

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order satisfactory to the Government, except that the operational inspection(4.9) shall be last.

Table II.- Group A Inspection.

Inspection	Rqt. Par.	Insp Par.	AQL Major	AQL Minor
Visual and mechanical	3.14	4.10		
Power Supply	3.14	4.10	1.0 dphu	4 dphu
R-F stage	3.14	4.10	1%	4%
R-F output attennator	3.14	4.10	1%	4%
Antenna	3.3.1.3	4.10	1.0 dphu	4 dphu
Transit Case	3.3.1.4	4.10	1%	4%
V & M over all(all sub-chassis mounted as required and inclusion of all spare parts and sub-components stored in transit case.)	3.14	4.10	1 dphu	4 dphu
Operating tests.-	3.11	4.9	* 1.5 dphu combined A Q L for all tests	
Input power	3.11.1	4.9.1		
Power switch	3.11.1.1	4.9.2		
R-F stage	3.11.3	4.9.1		
Power supply	3.11.2	4.9.3		
Sweep frequency	3.11.3.1	4.9.4 & 4.9.8		
R-F output attennation	3.11.3.2	4.9.5		
Modulation	3.11.4	4.9.1		
Antenna gain	3.11.5	4.9.6		
Audio power output	3.11.6	4.9.9		
Attennator pad(28 db)	3.11.7	4.9.7		

* All defects are considered major.

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4.5.2 Group B inspection.- This inspection, including sampling, shall conform to table III and the special procedures for small-sample inspection of MIL-STD-105. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

Table III- Group B Inspection

Inspection	Rqt. Par	Insp Par	A Q L (See 4.5.2.1)
Interchangeability	3.7	4.7	6.5
Immersion	3.8.5	4.8.4	6.5

4.5.2.1 Group B sampling Plan.- The group B sampling plan, for the A Q Ls listed in table III shall be L-7 for normal and tightened inspection. and L-5 for reduced inspection.

4.5.2.2 Order of inspection within group B.- Group B inspection shall be performed in any order satisfactory to the Government.

4.5.2.3 Procedure in case of failure.- When an inspection lot fails the contractor shall immediately investigate the cause of failure and take corrective action to assure that subsequent lots do not contain the same defect or defects.

4.5.3 Group C inspection.- This inspection shall be as listed in table IV and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

Table IV.- Group C inspection.

Inspection	Rqt Par	Insp Par
Bounce	3.8.10	4.8.8
Suppression of R-F interference	3.9	4.4

4.5.3.1 Sampling for inspection of equipment.- Three units for each group C inspection shall be selected from each 100 on order, without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

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4.5.3.2 Noncompliance.- If a sample unit fails group C inspection the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and the details of the corrective action taken on the process and all the units of the product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet the specified requirements, or if the contractor cannot determine the cause of the failure, the matter shall be referred to the contracting officer. (See 6.5).

4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on the contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.5.5 Disposition of nonconforming product.- The disposition of any nonconforming product shall conform to Standard MIL-STD-105. Such product shall be suitably tagged or identified to indicate the cause of failure and the means employed to correct the fault. The required information shall be presented to the Government when the product is submitted.

4.6. Bounce preconditioning.- The unit, less transit case, shall be placed in its normal operating position on the table of the Package Tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32- inch-thick strip of material under one corner or edge of the unit to a distance of three inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute. After bounce preconditioning the unit shall not be repaired , alined, cleaned or otherwise changed, prior to subjection to the acceptance inspection.

4.7. Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits it shall be considered a major defect.

(a) External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect the mating of parts.

(b) Dimensions of cavities when such dimensions affect the insertion of items.

(c) Location of hinges and fasteners on seperable parts or assemblies which must mate, such as cases, covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and the location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

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4.8 Service conditions.-

4.8.1 Temperature test.- The equipment shall be subjected to the temperature cycle shown in Standard MIL-STD-169. Performance shall be as follows:

Steps	Paragraph	Measurement	Performance
1, 4, 9 and 10	4. 9. 1	Modulated Output	Full specification performance.

4.8.2 Moisture resistance test.- The equipment shall be tested as follows:

4.8.2.1 Test conditions.-

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after the power is applied to the equipment. Complete the measurements as rapidly as possible. Do not leave the power on after the measurements have been completed.

(c) Test sealed equipment with the seal intact except during the last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against the entrance of moisture. Perform the last two cycles with the chassis removed from its enclosure.

4.8.2.2 Test method.-

(a) Dry at 130° ± 5°F for 24 hours.
(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.
(c) Take measurements as specified in 4.8.1 and readjust or realine as necessary to meet the full specification requirements.
(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.9.1 (a) shall be made during the periods shown in the standard and shall comply with 4.9.1(a).

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity. Then adjust the equipment for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment the equipment shall meet the full specified performance for those measurements specified in 4.9.1(a).

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4.8.2.3 Failure.- If any equipment fails to meet the performance specified in 4.9.1(a) during cycling it shall be realined or readjusted once. If the equipment then fails to meet the performance specified in 4.9.1(a), or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet the full specification requirements after conditioning and adjustment, as specified in 4.8.2.2.(e), it does not pass the test.

4.8.3 Elevation test.- The equipment shall be placed in an altitude chamber, in its normal operating position, and tested as follows: (Note: To check temperature stabilization place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the measurements of 4.9.1(a).

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level)and, after temperature and pressure stabilization, repeat the measurements specified in 4.8.3(a). The equipment shall meet the specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury(simulated 50,000 feet above sea level). Maintain this condition for one hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and after stabilization repeat the measurements specified in 4.8.3(a). The equipment shall meet the specified performance for these measurements.

4.8.4 Immersion test.- The equipment, as prepared for field transportation, shall be immersed to a depth of three feet of fresh water for two hours. Immediately prior to immersion the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ±2° F of its initial temperature or the temperature of the water shall be maintained within those limits by other means. After completion of the two-hour period of immersion the equipment shall be removed from the water and wiped dry on the exterior surfaces. There shall be no evidence of leakage when the equipment is opened.

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4.8.5 Rain test.- The equipment, minus the transit case, shall be tested as follows:

(a) Dry at 150° ± 5°F for 48 hours.

(b) Condition for four hours at 77° ± 5°F and 40 to 60 percent relative humidity.

(c) Take the measurements specified in 4.9.1(a) to insure that the equipment meets the specified performance.

(d) Place the equipment, in its normal operating position, in a test chamber maintained at 77° ± 5° F. Allow simulated rain to fall on exposed surfaces of the equipment. The water shall be at a temperature of 65° ± 5° F and shall fall freely at a rate of two inches per hour. Maintain this condition for one hour.

(e) With the rainfall continuing at the two-inch rate introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the equipment. The water shall fall on exposed surfaces and be blown by the wind against one side of the equipment. Maintain this condition for one hour.

(f) Rotate the equipment 90 degrees horizontally to expose a second side of the component to the wind and the rain. Maintain this condition for one hour.

(g) Continue as in(f) until all four sides of the equipment have been exposed to the wind and rain, each side for one hour.

(h) Immediately after completion of the foregoing repeat the measurements specified in (c) without removing the equipment from the chamber but with the wind and the rain discontinued. Complete the measurements within one hour. The equipment shall meet the specified performance.

(i) Open the equipment. There shall be no visible evidence of leakage.

4.8.6 Sand and dust test.- The equipment shall be tested as follows:

(a) Place the equipment in its normal operating position in the test chamber. Raise and maintain the sand and dust density at 0.1 to 0.5 gram per cubic foot within the test space. The relative humidity shall not exceed 30 percent at any time during the test.

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(b) The sand and dust shall be of angular structure and shall have a particle size such that the particles can pass through screens as follows:

Screen, U.S. Standard Sieve Series	Percent.
100- Mesh	100
140- Mesh	98 ± 2
200- Mesh	90 ± 2
325- Mesh	75 ± 2

(c) Chemical analysis of the sand and the dust shall be as follows:

Substance	Percent by weight
$Si\ O_2$	97 to 99
$Fe_2\ O_3$	0 to 2
$Al_2\ O_3$	0 to 1
$Ti\ O_2$	0 to 2
Mg O	0 to 1
Ignition losses	0 to 2

(d) Maintain the internal temperature of the test chamber at 77° ± 4° F for a period of six hours with the sand and dust velocity through the test chamber between 100 and 500 feet per minute.

(e) After six hours at the above conditions raise the temperature to 160° ± 4°F and maintain it at this value for six hours.

(f) Remove the equipment from the test chamber and allow it to cool to room temperature. There shall be no evidence of damage, other than surface abrasion, the equipment controls shall operate properly and the equipment shall meet the tests specified in 4.9.1(a).

4.8.7 Vibration test.- Interral vibration of the equipment shall be measured as follows: (See 3.8.9)

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. The mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate such observation and measurement, subassemblies may be tested seperately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

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(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.03 inch.

(c) In each of the three directions change the frequency in steps of one cycle per second and maintain each frequency for at least ten seconds.

(d) Measure vibration amplitude by optical means or by other means provided that vibration of the part is not affected by the measurement.

4.8.8 Bounce test.- The equipment shall be tested on a package tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal, as follows:

(a) Cover the tester bed with a panel of 1/2 -inch plywood with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at six-inch intervals around all four edges and at three- inch intervals in a six-inch square in the center.

(b) Place the equipment, in its transit case, on the bed of the package tester. Limit the lateral motion,by wooden fences, to not more than three inches and not less than one inch.

(c) Operate the package tester, shafts in phase, for a total of three hours at 283 ± 1 r. p.m. Turn the equipment 90 degrees at the end of each 45 minutes, each time in the same direction. Rest periods may be provided to permit cooling of the resilient mounts provided that the testing time is not less than three hours.

4.8.9 Shock test, bench handling.- The chassis and front panel assembly shall be removed from its enclosure and placed in a suitable position as for servicing, on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axis, for a total of four drops.

(b) Repeat "(a)", with the assembly resting on other faces until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

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4.8.10 Shock test: drop.- The equipment, in its transit case, shall be dropped, from a height of 24 inches, on each face, edge and corner, for a total of 26 drops. The floor or barrier receiving the impact shall be of two-inch fir backed by concrete or a rigid steel frame. Drops shall be made from a quick release hook, or pendulum tester as made by the L. A. B. Corporation, Skaneateles, New York, or equal.

4.9 Operating tests.- The test equipments used, to perform the tests listed herein, shall be selected from the following list or a substitute may be used, as approved by the contracting officers' technical representative:

Detector- Amplifier assembly	AN/UPA-1( )
Signal Generator	TS- 202( )/U
Signal Generator	AN/URM - 64 ( )
Generator, Signal	AN/URM - 33 ( )
Spectrum Analyzer	AN/UPM- 17 ( )
Frequency Meter	TS -186 ( )/U P
Wavemeter	FR -49 ( )/U
Wavemeter	FR-53 ( )/U
Wave and Power Meter Set	TS-107U/TPM-1
Wattmeter	ME-69 ( )/U
Multimeter	TS-505 ( )/U
Multimeter	ME-6 ( )/U
Multimeter	ME- 30 ( )/U
Audio Oscillator	TS -382( )/U.

4.9.1 Input power vs R-F output tests.-

(a) The input power source shall be adjusted to 115 volts, 60 cycles per second(cps) and the R-F output measured for frequency, power and amplitude stability. The R-F output shall be measured at 1220, 1250, 1285, 1300 and 1350 mc while transmitting CW, sine wave, square wave, pulse, and noise modulated signals.

(b) Repeat the above measurements with the input power adjusted to 105, 110, and 120 volts.

(c) Adjust the input power to 115 volts, 48 cps, and repeat all measurements.

(d) Repeat all measurements with the input power frequency at 62 cps, 115 volts.

(e) Repeat all measurements with the input power frequency adjusted to 390 cps, 400 cps, 420 cps, and 440 cps.

4.9.2 Power switch test.- At the start of the tests of 4.9.1 the power switch shall be tested for compliance with 3.11.1.1.

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4.9.3 Power supply tests.- When the entire Trainer, Radar AN/ULT-T4( ) is being procured the power supply unit need not be tested as a seperate unit, the tests of 4.9.1 will be sufficient. When the power supply is being procured as a seperate unit the power supply outputs shall be loaded as specified in 3.11.2 and voltage measurements made to determine compliance with the voltage requirements of 3.11.2.

4.9.3.1 Fusing tests. -During any R-F output test, or power supply output test, the fuses shall be removed from the circuits and appropriate voltage measurements made to determine compliance with the requirements of 3.11.2.1.

4.9.4 Sweep frequency test.- At the conclusion of the test of 4.9.1(a) the sweep frequency requirement of 3.11.3.1 shall be tested for.

4.9.5 R-F output attennation.- When the entire Trainer, Radar AN/ULT-T4( ) is being procured the R-F output attennation shall be tested during the tests of 4.9.1. If the output attennator is being procured seperately it shall be connected to a signal generator and the attennator output measured for compliance with the requirements of 3.11.3.2.

4.9.6 Antenna gain tests.- When the entire Trainer, Radar AN/ULT-T4( ) is being procured the antenna gain shall be measured with Detector-Amplifier Assembly AN/UPA-1 ( ), or equal. When the antenna is being procured seperately the gain shall be measured by feeding the antenna with an approved signal generator and measuring the radiated power with the AN/UPA-1( ), or equal.

4.9.7 Attennator pad, 28 db, tests.- The attennator pad, 28 db, shall be tested at its input and output, while in its circuit, by measuring its input and output power. When procured seperately it shall be tested in a like manner, but with an approved signal generator as the input source.

4.9.8 Calibration accuracy tests.- During the tests of 4.9.1 the calibration accuracies of the sweep, sweep rate, and variable frequency controls shall be checked.

4.9.9 Audio Power output test.- The audio power output shall be measured, using a multimeter or power meter, for compliance with the requirements of 3.11.6.

4.10. Visual and mechanical inspection.- The equipment shall be examined for the defects listed in Standard MIL-STD-252 and Table V:

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Table V.- Classification of visual and mechanical defects.-

Classification	Defect
Major	Carrying handle, hinge, or lid fasteners attached to wrong side of case. Cover missing or inoperative. Control inoperative. Units do not fit in transit case.
Minor	Cover or case reinforcement missing or improperly assembled. Control difficult to operate. Difficulty in obtaining good contact or in making the required setting of a control due to play or backlash. Carrying handle, cover or catches difficult to operate.

4.11 Acceptance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Signal Corps drawing SC-A-362100.

5. PREPARATION FOR DELIVERY

5.1. Trainer, Radar AN/ULT-T4( ) shall be preserved, packaged, packed, marked and tested in accordance with the applicable provisions of Drawing SC-A-362100.

6 NOTES

6.1 Intended use.- Trainer, Radar AN/ULT-T2 is used to introduce interference signals into a radar receiver, necessary to instruct student radar operators in anti-jamming procedures.

6.2. Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment.(Level A, level B or level C.)

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(c) Preproduction inspection:

(1) Two sample units of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit. (See 3.4)

(2) Preproduction pack(s) as follows:
Makeup of pack(s)
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required.(See 3.13)

(g) Quantity of tools and running spare parts required.(See 3.13)

6.3. Color.- The color chip furnished by the procuring agency (See 3.5) will be color chip Nr. X-24087 and may be obtained upon request to the Commanding General U. S. Army Signal Supply Agency, 225 South Eighteenth Street, Philadelphia, 3, Pennsylvania, ATTN: SIGSU-J4b.

6.4. Definition of level B.- Level B packaging requirements, detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments, covered storage for a period of three years or less and distribution within the continental limits of the United States.

6.5. Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending a decision from the contracting officer on the adequacy of the corrective action.

6.6. Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: AM/ULT-T2W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.7. Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

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(a) Type A- The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B- That inspection set forth in the Quality Assurance Provisions of this specification, or the contract, reduced in amount under the reduced inspection provisons of Military Standard 105.

(c) Type C- A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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